UNIVERSITY OF OXFORD
SOFTWARE ENGINEERING PROGRAMME

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Part-time postgraduate study in software engineering

Classical Machine Learning, CML

26th – 30th January 2026

ASSIGNMENT

The purpose of this assignment is to test the extent to which you have achieved the learning objectives of the course. As such, your answer must be substantially your own original work. Where material has been quoted, reproduced, or co-authored, you should take care to identify the extent of that material, and the source or co-author.

Your answers to the questions on this assignment should be submitted using the Software Engineering Programme website — `www.softeng.ox.ac.uk` — following the submission guidelines. When submitting the assignment online, it is important that you formally complete all three assignment submission steps: step 1, read through the declaration; step 2, upload your files; step 3, check your files. **Please ensure your submission is anonymous: do not include your name or any other identifying information on the assignment, nor in accompanying material such as source code, nor within the file names of anything submitted.**

The deadline for submission is 12 noon on Tuesday, 17th March 2026. You are strongly encouraged to submit a version well before the deadline. You may update your submission as often as you like before the deadline, but no submissions or changes will be accepted after the deadline.

We hope to have preliminary results and comments available during the week commencing Monday, 27th April 2026. The final results and comments will be available after the subsequent examiners' meeting. Exam Conventions can be found here `https://www.softeng.ox.ac.uk/handbook/`

ANY QUERIES OR REQUESTS FOR CLARIFICATION REGARDING THIS ASSIGNMENT OR PROBLEMS INSTALLING SOFTWARE SHOULD, IN THE FIRST INSTANCE, BE DIRECTED TO THE PROGRAMME OFFICE WITHIN THE NEXT TWO WEEKS.

Classical Machine Learning (CML) Jan 2026

1 Assignment Task

1.1 Overview

This assignment is focussed on modelling continuous variables. In particular it is concerned with 'understanding' and predicting relationships between data-series.

There are two parts to the task:

1. Direct modelling of three datasets; and
2. Explanation and description of the modelling algorithms in a tutorial style.

The assignment is intended to test both practical ability in data analysis and also theoretical understanding of the algorithms and methods that are used in this domain.

1.2 Data

You have been provided with three data-files:

1. 'dataset_1.csv';
2. 'dataset_2.csv'; and,
3. 'dataset_3.csv'.

As the naming implies, these are comma-separated datasets containing columns of numerical data.

In these datasets, input features (independent variables) are denotated with a 'X', whilst output ('Target', 'Response' or 'Dependent') variables are denoted with a lower-case 'y'. In the case of 'dataset_2.csv', there are two independent variables 'X1' and 'X2'.

1.3 Specific Tasks

1.3.1 Task 1

Model the relationship between the input features and response variables using a variety of models. For each of the models you use, evaluate the performance of the model in qualitative and quantitative terms. Consider both predictive performance, computational efficiency of the algorithm and other evaluation criteria that are relevant. Describe the advantages and disadvantages of each algorithm.

The goal is to build one or models that perform well for each of the datasets.

Include succinct explanatory comments and rationale in 'markup' cells that will enable a reader to follow the overall logic of your analysis, results and conclusions. Document any assumptions you make.

1.3.2 Task 2

Select one, two or three of the models you developed in Task 1 and explain the algorithms used to generate the models. Document your explanation in the form of a tutorial that is suitable for other students studying the Classical Machine Learning (CML) module.

Note:

1. If you decide to adopt the style of presentation used in the course 'CML Workbook' then you are required to include all code explicitly. That is, you are required to deliver a 'Solution' workbook not a 'Problem' workbook;
2. The task is intended to test your understanding of the algorithms and also your ability to express and explain these in a comprehensible and useful manner;
3. No distinction will be made in the marking regarding the number of algorithms you choose to describe (1, 2 or 3). You might equally decide to provide an extensive tutorial on a single algorithm; to develop a tutorial the compares and contrasts up to a maximum of three algorithms or simply to provide 2 or 3 largely independent tutorial sections.

2 Submission Requirements

2.1 Use Python in a Jupyter Notebook

Develop your models within two Jupyter Notebooks – one notebook for each task.

Your analysis and modelling is to be completed using the Python Language. You must include a section within your submission that identifies which libraries you have used, with references identifying their sources.

2.2 Heading Numbering and Table of Contents

It is recommended that you:

1. Number headings and sub-headings in your Jupyter Notebook;
2. Include a table of contents of all headings and sub-headings near the start of your notebook

Such numbering helps structure your assignment and makes it more navigable and readable. It will enable the examiner to provide feedback that is easier to reference back to your submission.

Add-on tools for VSCode that enable heading numbering and generation of table of contents are available at https://marketplace.visualstudio.com/ .

2.3 Character Count

There is a 'character count' limit of 75,000 characters for your submission. That is, the number of characters included within your Python code and your explanatory markdown cells must be less than a combined total of 75,000. This limit does not include text appearing in output from your code.

Include the cell shown in Figure 1 at the start of your Jupyter Notebook. Change the filename passed to the 'count_chars' function to match the file name of your final, submitted .ipynb file. As a final action before submission, execute this code. Include its output in both the submitted .ipynb file and the exported PDF version.

If for this code does not run in your environment then you may modify the code within reason to make it work. If you need to do this, then you must provide an explanation of any modifications you make. The examiner will re-run the version of the code shown in Figure 1 on each submission. The character count obtained by the examiner will be used as the final arbiter of submission length.

Do not attempt to ‘game’ this metric; you should adopt function and variable naming conventions and documentation that suits the task at hand rather than adopting poor conventions because of the character count limit.

```python
import json

def count_chars(filename):
    with open(filename) as f:
        nb = json.load(f)

    counts = {'code': 0, 'markdown': 0, 'raw': 0, 'other': 0}

    for cell in nb['cells']:
        cell_type = cell['cell_type']
        chars = sum(len(line) for line in cell['source'])
        if cell_type in counts:
            counts[cell_type] += chars
        else:
            counts['other'] += chars

    total = sum(counts.values())

    for cell_type in ['code', 'markdown', 'raw', 'other']:
        print(f"{cell_type},{counts[cell_type]}")

    print(f"total,{total}")

    if total > 75000:
        print("Too many characters in assignment submission (> 75,000)")
    else:
        print("Number of characters ok ( < 75,000)")

count_chars('count_chars_in_ipynb.ipynb')
```

```
code,810
markdown,66
raw,0
other,0
total,876
Number of characters ok ( < 75,000)
```

Figure 1: Code to Count number of Characters in a Jupyter Notebook

2.4 Submit in both .ipynb and PDF format

You are required to submit **two** versions of any Jupyter Notebooks you submit:

- A fully-executed version of .ipynb file (that is, containing all outputs) : so that the examiner can execute the code if required;
- A **pdf version** of the fully-executed notebook : if the examiner cannot execute any cell within your .ipynb file it will still be clear how that code operated on your machine

2.5 Constraints

Do not include any analysis employing members of the family of Neural Networks (including large language models).

Do not include time-series analysis models (E.g. Autoregressive (AR) models such as ARIMA / SARIMA; Impulse response models). Any such analysis is outside of the scope of the course and hence of this assignment and will not attract any credit.

Be succinct and focus on the specific tasks as stated. Avoid digressions and over-elaboration of the tasks.

Do not use Large Language Models (LLMs) or 'co-pilots' to perform analysis, generate code or supporting rationale.

Ensure that any cross-referencing is clear and unambiguous.

3 Assessment Criteria

This assignment is intended to assess your ability to:

- Employ the tools and techniques presented as part of the Classical Machine Learning course succinctly and efficiently in novel situations;
- Make critical judgments regarding the applicability and suitability of Classical Machine Learning algorithms to specific problems;
- Provide rationale, explanation and documentation for any decisions you make whilst engaged in a Classical Machine Learning project; and,
- Critically assess the quality and limitations of any models you build and/or analysis that you complete.

Appendix A : Checklist to support your review of your submission

This appendix is not part of the formal definition of this assignment – the authoritative assignment text appears in the main body. However, it is intended to support your submission and to act as an aid to navigation of the above requirements and constraints.

#	Description	Done?
1	Included a .ipynb file containing the final submission?	
2	The .ipynb file has been fully executed and the output is available in the file?	
3	Included a PDF file containing the final, executed version of your submission?	
4	Included a cell that lists all of the libraries used in your code and their sources?	
5	(Recommended) Each heading and sub-heading numbered?	
6	(Recommended) Included a table of contents?	
7	‘count_chars’ function run on the final submitted version of your code?	
8	Resulting output from ‘count_chars’ function visible in the output?	
9	Character count <75,000 characters?	